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A note on online strip packing

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Abstract In online strip packing we are asked to pack a list of rectangles one by one into a vertical strip of unit width, without any information about future rectangles. The goal is to minimize the total height of strip used. The best known algorithm is *First Fit Shelf algorithm* (Baker and Schwarz in SIAM J. Comput. 12(3):508–525, 1983), which has an absolute competitive ratio of 6.99 under the assumption that the height of each rectangle is bounded from above by one. We improve the shelf algorithm and show an absolute competitive ratio of $7/2 + \sqrt{10} \approx 6.6623$ without the restriction on rectangle heights. Our algorithm also beats the best known online algorithm for parallel job scheduling.

Keywords Strip packing · Online algorithms · Competitive ratio

1 Introduction

We study a two-dimensional strip packing problem, in which a number of rectangles with widths bounded by 1 are packed into a strip with width 1 and infinite height.

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The rectangles must be packed such that no rectangles overlap and the sides of the rectangles are parallel to the strip sides. Rotations are not allowed. The objective is to minimize the height of the strip used to pack all the rectangles. Strip packing arises in many real-world applications, e.g., VLSI design, stock cutting problem, for packing newspaper commercials. This problem was proved to be NP-hard (Baker et al. 1980) in 1980. In this paper we consider the *online* version of this problem, in which rectangles are arriving one by one and a placement decision for the current rectangle must be made before the next rectangle is given. Furthermore, once a rectangle has been packed into the strip, we cannot change its placement later.

Competitive ratios To evaluate an online algorithm we adopt the standard measure—*competitive ratio* (Borodin and El-Yaniv 1998). For any input list L of items and an online algorithm A, we denote by $OPT(L)$ and $A(L)$, respectively, the height of the strip used by an optimal algorithm and the height of the strip used by algorithm A to pack list L. The *absolute competitive ratio* of A, denoted by R_A, is given by

$$R_A = \sup_L \{A(L)/OPT(L)\}.$$

The *asymptotic competitive ratio* R_A^∞ of algorithm A is defined by

$$R_A^\infty = \lim_{n\to\infty} \sup_L \max\{A(L)/OPT(L) \mid OPT(L) = n\}.$$

Related work Strip packing has been extensively studied. Coffman et al. (1980) presented the algorithms NFDH (Next Fit Decreasing Height) and FFDH (First Fit Decreasing Height) for strip packing and showed that their asymptotic performance ratios are 2 and 1.7, respectively. An asymptotic fully polynomial time approximation schemes (AFPTAS) were given by Kenyon and Rémila (1996). Recently Jansen and Stee (2005) dealt with the case that rotations of 90° are allowed and presented an AFPTAS. Regarding the absolute performance ratio, the best known algorithms have an upper bound of 2 (Schiermeyer 1994; Steinberg 1997).

Concerning the online strip packing problem, previous results mainly focus on asymptotic competitive ratios. Baker and Schwarz (1983) developed the first online algorithm, the shelf algorithm, with an asymptotic competitive ratio close to 1.7. The best shelf packing algorithm with asymptotic competitive ratio 1.691 was shown by Csirik and Woeginger (1997). Very recently Han et al. (2007) explored the relation between strip packing and one dimensional bin packing and thus showed that strip packing admits an online algorithm with asymptotic ratio of 1.5888.

A lower bound of 2 for the absolute competitive ratio of online strip packing was given by Brown et al. (1982). Baker and Schwarz (1983) showed that the first fit shelf algorithm (FFS) for online strip packing has absolute competitive ratio of 6.99. To our best knowledge, it is the current best upper bound for online strip packing problem. Moreover, as stressed by Csirik and Woeginger (1998), the bound assumed that the height of any rectangle is at most 1.

A closely related problem is parallel job scheduling (Du and Leung 1989; Drozdowski 2004; Johannes 2006; Ye and Zhang 2007). A parallel job can be executed on more than one processor simultaneously. If we regard the widths as the resource

(processors) and the heights as the time, strip packing becomes a parallel job scheduling. The only difference is that in parallel job system the processors required by a job are not necessarily consecutive, while in strip packing the geometrical structure of processors will be considered. Any algorithm for strip packing applies to parallel job scheduling (Johannes 2006; Ye and Zhang 2007). Johannes (2006) gave a lower bound of 2.25 with an enumerating program. Hurink and Paulus (2007a) recently improved the lower bound to 2.43. Note that the lower bounds in (Johannes 2006; Hurink and Paulus 2007a) are valid for the strip packing problem. Ye and Zhang (2007) presented an online algorithm with a competitive ratio of 7.

For the case with two processors Chan et al. (2007) showed that no online algorithm can achieve an absolute competitive ratio less than 1.816. Hurink and Paulus (2007a) proved a tight bound of 2.

Our contribution We consider the absolute competitive ratio for online strip packing. By revisiting the shelf algorithm we present a simple heuristic and show an improved absolute competitive ratio of $3.5 + \sqrt{10} \approx 6.6623$. Note that in our algorithm the height of any rectangle can be an arbitrary number, i.e., we do not use any restriction on the height of rectangles.

2 An improved online algorithm

In this section we propose an online algorithm with absolute competitive ratio at most 6.6623. We also provide an instance to show that the upper bound is tight. The idea is to apply shelf packing to grouped rectangles.

We re-state the online strip packing problem as follows: Given a list of rectangles $I_j = (w_j, h_j)$, $j = 1, 2, \ldots, n$ and $0 < w_j \leq 1$, where each rectangle is characterized by its width w_j and its height h_j. The rectangles arrive over list, which are not known in advance. Upon arrival of rectangle I_j, its width and height become known, and it must immediately and irrevocably be assigned to the strip. The next rectangle I_{j+1} appears only after I_j has been assigned. Our goal is to pack all the rectangles into a vertical strip of unit width to minimize the height of strip used. Rectangles are not allowed to rotate or overlap.

In a shelf packing, basically, the strip is gradually divided level by level. The first level is the bottom of the bin and each new level is defined by drawing a horizontal line across the strip with a given height above the previous level. A shelf is the rectangular bin formed by two consecutive levels and its height gives the distance of the two levels.

Before describing our algorithm, we group rectangles into two classes. A rectangle (w, h) is called *big* if $w > 1/2$, and *small* otherwise. Let $0 < r < 1$.

Algorithm Revised-Shelf (RS_r):

As a rectangle $I_i = (w_i, h_i)$ arrives, we do packing as follows:

1. If rectangle I_i is *big*, i.e. $w_i > 1/2$. Open a new shelf with height of h_i and place I_i in.

2. If rectangle I_i is *small*, i.e., $w_i \leq 1/2$, pick an integer k (can be negative) such that $r^{k+1} < h_i \leq r^k$. Pack I_i into a shelf of height r^k with the FF (First Fit) algorithm. More precisely, the rectangle I_i is packed leftmost to a lowest shelf with height r^k that has enough room for I_i. Otherwise, create a new shelf with height r^k at the current top of the strip for I_i.

Different from the standard shelf algorithm we apply shelf packing to grouped rectangles and we allow the power k to be negative. The former helps us improve the competitiveness and the latter allows an arbitrary height of a rectangle.

Let us take a look at the final packing by algorithm RS_r. It consists of shelves. We group the shelves into three classes. Those created for *big* rectangles are called *full* shelves. For each k, the last one of height r^k created for *small* rectangles is called *sparse* shelf and the remaining shelves are *dense*. Denote H_F to be the total height of *full* shelves, H_S to be the total height of the *sparse* shelves, and H_D to be the total height of the *dense* shelves, respectively.

Theorem 1 *The absolute competitive ratio of algorithm RS_r is at most* $\max\{\frac{3}{2r}, 2\} + \frac{1}{r(1-r)}$, *the minimum of which is* $7/2 + \sqrt{10} \approx 6.6623$ *by setting* $r = \frac{5-\sqrt{10}}{3}$.

Proof We first consider the *sparse* shelves. Let h be the height of a highest *sparse* shelf and H the height of the tallest rectangle in all *sparse* shelves. Since in each shelf, the height of each rectangle is greater than r times the height of the shelf, we have $H > rh$. For any $k \in \mathbb{Z}$, as there is at most one *sparse* shelf of height r^k, we get an estimating upper bound of H_S as follows (recall that h is the largest height among the *sparse* shelves. The following sum thus starts with index zero)

$$H_S \leq h \sum_{i=0}^{\infty} r^i = \frac{h}{1-r} < \frac{H}{r(1-r)}.$$

We then want to prove that the total area of the rectangles packed in *dense* shelves and *sparse* shelves is at least $2r \cdot H_D/3$. Consider the *dense* shelves with height r^k. Let n_k be the number of these shelves. If $n_k = 0$, then the dense shelves of height r_k has no contribution to H_D and our claim trivially holds. Assume that $n_k \geq 1$. Then there must be a *sparse* shelf of the same height. We deal with the following three cases.

Case 1. $n_k = 1$. Denote the *dense* shelf to be s. Since any rectangle in the *sparse* shelf does not fit in s, we have the total width of the rectangles in shelf s and the *sparse* shelf is greater than one. Thus the total area of them is more than r^{k+1}. It suffices for our claim.

Case 2. $n_k = 2$. Let s_1 and s_2 be the two *dense* shelves. Since in every *dense* shelf the total area of rectangles packed is at least $r^{k+1}/2$, together with the argument of Case 1, we have the total area of rectangles in s_1 and s_2 and the sparse shelf is at least $\frac{3r^{k+1}}{2}$. Note that these *dense* shelves contribute $2r^k$ to H_D. The claim holds.

Case 3. $n_k \geq 3$. A *dense* shelf contains at least two (small) rectangles. Let s_1, s_2, s_3 be any three *dense* shelves of height r^k. W.l.o.g., we suppose that s_3 is created after s_1 and s_2. Any rectangle in s_3 cannot fit in either s_1 or s_2. It implies that the total

width of rectangles packed in s_1, s_2 and s_3 is more than 2. The total area of them is thus larger than $2r^{k+1}$. The above argument shows that the average area of rectangles in each dense shelf is larger than $2r^{k+1}/3$. It shows that a shelf contributes r^k to H_D and more than $2r^{k+1}/3$ to the area. It gives the claim.

Finally, we consider the *full* shelves. Clearly, the total area of rectangles packed in these shelves is at least $H_F/2$.

Let ALG, OPT be the height of strip reached by algorithm RS_r and an optimal offline algorithm, respectively. Then $ALG = H_F + H_D + H_S$. To estimate OPT it is easy to see that any optimal algorithm cannot use a strip of height less than either the total area of rectangles or the largest height among all the rectangles. Thus we have

$$OPT \geq (H_F/2 + 2rH_D/3) \geq \min\{1/2, 2r/3\}(H_F + H_D)$$

and

$$OPT \geq H > r(1-r)H_S.$$

Then

$$R_{RS_r} = \frac{ALG}{OPT} \leq \max\left\{\frac{3}{2r}, 2\right\} + \frac{1}{r(1-r)}.$$

As $r = (5-\sqrt{10})/3$, R_{RS_r} arrives at the minimum value of $7/2 + \sqrt{10}$. □

Theorem 2 *For any* $0 < r < 1$, *the absolute competitive ratio of algorithm* RS_r *is at least* $\frac{3}{2r} + \frac{1}{r(1-r)}$, *which is at least* $7/2 + \sqrt{10} \approx 6.6623$ *achieved by setting* $r = \frac{5-\sqrt{10}}{3}$.

Proof Let $H = r + \varepsilon$, and k be a sufficiently large positive integer. Let $n = \lceil \frac{H}{r^k+\varepsilon} \rceil$, without loss of generality, we assume n to be even. For any given $0 < r < 1$, choose ε, δ so that $0 < \varepsilon < r^k - r^{k+1}$ and $0 < \delta < 3\varepsilon/k$. Let $\Delta > \varepsilon$. Based on the bad example for the First Fit bin packing algorithm (Johnson et al. 1974), an instance is constructed as follows:

The first $k-1$ rectangles arrive in the order of $(\delta, r^{k-1}+\varepsilon), \ldots, (\delta, r^2+\varepsilon), (\delta, r+\varepsilon)$ (we say these rectangles are of type Z). Then $3n-1$ rectangles of the same height get released as $\{b_{n-1}, a_{1,n}, a_{2,n}, b_{n-2}, a_{1,n-1}, a_{2,n-1}, \ldots, b_1, a_{1,2}, a_{2,2}, a_{1,1}, a_{2,1}\}$, where

$$b_j = (1/3 - 2^{2j+1}\Delta - \varepsilon, r^k + \varepsilon), \quad j = 1, 2, \ldots, n-1,$$

$$a_{1,j} = a_{2,j} = (1/3 + 2^{2j}\Delta - \varepsilon, r^k + \varepsilon), \quad j = 1, 2, \ldots, n.$$

One can see that all the rectangles can be packed into the strip of height $H + r^k + \varepsilon$. More precisely we can pack the rectangles of Z one next to another at the bottom right of the bin, using a total width of $(k-1)\delta$ and a height of H. The triple $\{b_j, a_{1,j}, a_{2,j}\}$ can be packed together in a row starting at the bottom left of the strip for $j = 1, 2, \ldots, n-1$, while the last two rectangles $a_{1,n}, a_{2,n}$ are placed in a row on the top of the strip. There are in total n rows, the total height of which is $n(r^k + \varepsilon) < H + r^k + \varepsilon$.

Now we turn to algorithm RS_r. Note that all rectangles are small. The algorithm runs exactly the same as the First Fit Shelf algorithm. The rectangles of Z need a height arbitrarily close to $H/(r(1-r))$ as k goes to infinity, since each rectangle of Z with height $r^j + \varepsilon$ takes by itself a shelf of height r^{j-1} for $j = 1, 2, \ldots, k-1$.

Next we consider the last $3n-1$ rectangles. Each rectangle b_j is followed by rectangles $a_{1,j+1}$ and $a_{2,j+1}$. One can check that with the First Fit Shelf algorithm exactly $3n/2$ shelves of height r^{k-1} are created. If we re-number the rectangles as $c_1, c_2, \ldots, c_{3n-1}$, then c_{2j-1} and c_{2j} go to a shelf for $j = 1, 3n/2-1$ and the last rectangle gets a shelf alone. The total height of these shelves is $3nr^{k-1}/2$, which tends to $3H/(2r)$ as ε goes to zero. Thus the total height of algorithm RS_r uses is arbitrarily close to

$$\frac{3H}{2r} + \frac{H}{r(1-r)}.$$

We have

$$R_{RS_r} \geq \frac{3}{2r} + \frac{1}{r(1-r)}.$$

The right-hand reaches the minimum value 6.6623 when $r = (5-\sqrt{10})/3 \approx 0.61257$. □

Combining with Theorem 1 we eventually prove a matching bound of algorithm RS_r.

3 Conclusions

In this paper we studied two-dimensional online strip packing problem. By adopting a novel idea that combines grouping rectangles and shelf packing, we reduced the absolute competitive ratio to 6.6623 for the online strip packing problem. We very recently noted that the same result was independently obtained by Hurink and Paulus (2007b). Some interesting problems remain in the absolute competitive analysis. One direction worthy studying is narrowing the gap of absolute competitive ratios [2.43, 6.6623] for online strip packing. Another topic is to look at multi-dimensional strip packing.

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